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Brigham Young University, Provo, UT, USA
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PROFESSIONAL EXPERIENCE

Professor, Brigham Young University, July 2024-present.

Visiting Professor, Bandung Institute of Technology, January-June 2025.

Graduate Coordinator, Mathematics Department, Brigham Young University, July 2021-July 2024.

Associate Professor, Brigham Young University, July 2019-July 2024.

Assistant Professor, Brigham Young University, October 2013-July 2019.

Visiting Assistant Professor, Tulane University, January-May 2018.

Postdoctoral Research Associate, Los Alamos National Laboratory, February 2012-October 2013 .

ACADEMIC PREPARATION

Ph.D. Applied and Interdisciplinary Mathematics University of Michigan, Jan. 2012.

B.S. Mathematics, Brigham Young University, April 2006.

JOURNAL PUBLICATIONS

(graduate student, and undergraduate student)

- (1) *Embracing Uncertainty in “Small Data” Problems: Estimating Earthquakes from Historical Anecdotes*, N. E. Glatt-Holtz, R. A. Harris, A. J. Holbrook, J. Krometis, **Y. Kurniawan**, **H. Ringer**, and JPW, Journal of Geophysical Research: Machine Learning and Computation (2025).
- (2) *Model discovery on the fly using continuous data assimilation*, **J. Newey**, JPW, and E. Carlson, Journal of Computational Physics (2025).
- (3) *Relaxation-based schemes for on-the-fly parameter estimation in dissipative dynamical systems*, V. R. Martinez, **J. Murri**, and JPW, Inverse Problems (2025).
- (4) *eGAD double descent is explained by Generalized Aliasing Decomposition*, M. K. Transtrum, G. L. W. Hart, T. J. Jarvis, JPW, Physical Review Research (2025).
- (5) *Error propagation of direct pressure gradient integration and a Helmholtz-Hodge decomposition based pressure field reconstruction method for image velocimetry*, **L. Li**, J. McClure, G. B. Wright, JPW, J. Wang, Z. Pan, Experiments in Fluids (2025).
- (6) *A tale of two faults: Statistical reconstruction of the 1820 Flores Sea earthquake using tsunami observations alone*, **T. Paskett**, JPW, R. A. Harris, **C. Ashcraft**, J. A. Krometis, **I. Sorensen**, and **R. Wonnacott**. Geo. J. Int, 237 (1), 419–435 (2024).
- (7) *Identifying the body force from partial observations of a 2D incompressible velocity field*, A. Farhat, A. Larios, V. Martinez, and JPW Phys. Rev. Fluids, 9 (5), (2024).
- (8) *Hydrodynamic stability in the presence of a stochastic forcing: convection as a case study*, JPW, J. Földes, N. Glatt-Holtz, and G. Richards. Physica D. (2024).

- (9) *A Simple Boundary Condition Regularization Strategy for Image Velocimetry Based Pressure Field Reconstruction*, **C. Pryce**, **L. Li**, JPW, and Z. Pan. Exp. in Fluids (2024).
- (10) *Imbricated Beachrock Deposits Adjacent to the Java Trench, Indonesia: Influence of Tsunami and Storm Waves*, R. A. Harris, **W. Meservy**, **H. Sulaeman**, M. Bunds, J. Andreini, **B. Sharp**, **B. Berrett**, JPW, **G. Carver**, G. Setiadi, S. Hapsoro, C. Prasetyadi. Nat. Haz. (2023).
- (11) *Error propagation dynamics of PIV-based pressure field calculation (3): what is the minimum resolvable pressure in a reconstructed field?*, **M. Nie**, JPW, B. L. Smith, and Z. Pan, Exp. in Fluids, 63 (11), 1–26 (2022).
- (12) *Concurrent multi-parameter learning demonstrated on the Kuramoto-Sivashinsky equation*, **B. Pachev**, JPW, and **S. A. McQuarrie**, SIAM Sci Comp., 44 (5), (2022).
- (13) *Dynamically learning the parameters of a chaotic system using partial observations*, E. Carlson, J. Hudson, A. Larios, V. R. Martinez, **E. Ng**, and JPW, Disc. and Cont. Dyn. Sys., (2021).
- (14) *Error propagation dynamics of velocimetry-based pressure field calculations (2): on the error profile*, **M. Faiella**, **C. G. J. Macmillan**, JPW, and Z. Pan. Meas. Sci. and Tech., 32 (8), 084005 (2021).
- (15) *Methodological reconstruction of historical seismic events from anecdotal accounts of destructive tsunamis: a case study for the great 1852 Banda arc mega-thrust earthquake and tsunami*, **H. Ringer**, JPW, J. Krometis, R. A. Harris, N. Glatt-Holtz, **S. Giddens**, **C. Ashcraft**, **G. Carver**, **A. Robertson**, **M. Harward**, **J. Fullwood**, **K. Lightheart**, **R. Hilton**, **A. Avery**, **C. Kesler**, **M. Morrise**, and **M. H. Kein**. J. of Geo. Res.: Solid Earth, 126 (4) (2021).
- (16) *Algebraic bounds on the Rayleigh-Bénard attractor*, **Y. Cao**, M.S. Jolly, E. S. Titi, and JPW. Nonlinearity, 34 (1) 509 (2021).
- (17) *Exact relations between Rayleigh Bénard and rotating plane Couette flow in 2D*, B. Eckhardt, C. R. Doering, and JPW. J. Fluid Mechanics RAPIDS, 903 R4 (2020).
- (18) *Rigorous bounds on the heat transport of rotating convection with Ekman pumping*, **B. Pachev**, JPW, G. Fantuzzi, and I. Grooms. J. of Math. Phys., 61 (2), 023101 (2020).
- (19) *Data Assimilation in Large Prandtl Rayleigh-Bénard Convection From Thermal Measurements Only*, A. Farhat, N. Glatt-Holtz, V. Martinez, **S. McQuarrie**, and JPW. SIAM J. of App. Dyn. Sys. 19 (1), 510-540 (2020).
- (20) *An Experimental Investigation of Kelvin-Helmholtz Instability in Separated Blood*, **R. L. Wood**, JPW, **A. K. Hunter**, **D. S. McClellan**, and W. G. Pitt. Am. Inst. of Chem. Eng. J. 65 (4), 1376-1386 (2019).
- (21) *Boundary layer analysis for Navier-slip Rayleigh-Bénard convection: the nonexistence of an ultimate state*, G.-M. Gie, and JPW. J. Math. Fluid Mech., Vol 21 (1), 3 (2019).
- (22) *The effect of two distinct fast time scales in the rotating, stratified Boussinesq equations: Variations from Quasi-Geostrophy*, JPW, T. A. Haut, and B. A. Wingate. Theo. & Comp. Fluid Dyn., Vol 32 (6), 713-732 (2018).
- (23) *Error propagation dynamics of PIV-based pressure field calculations: How well does the pressure Poisson solver perform inherently?* **Z. Pan**, JPW, S. Thomson, and T. Truscott. Meas. Sci. and Tech., Vol 27 (8), 14 pages (2016).

- (24) *Ergodicity in Randomly Forced Rayleigh-Bénard Convection.* J. Földes, N. Glatt-Holtz, G. Richards, and JPW. *Nonlin.*, Vol 29 (11), 3309-3345 (2016).
- (25) *Stability of vortex solutions to an extended Navier-Stokes system.* G.-M. Gie, C. Henderson, G. Iyer, L. Kavlie, and JPW. *Comms. in Math. Sci.*, Vol 14 (7), 1773-1797 (2016).
- (26) *Bounds on heat transport in Rapidly Rotating Rayleigh-Bénard Convection.* I. Grooms and JPW. *Nonlinearity*, Vol. 28, 29–41 (2015).
- (27) *Potential Vorticity: a diagnostic tool for General Circulation Models.* JPW, C. Jablonowski, J. Kent, and R. B. Rood. *Quart. J. of the Roy. Met. Soc.*, Vol. 141 (688), 739–751 (2015).
- (28) *The influence of fast waves and fluctuations on the evolution of the ‘slow manifold’.* JPW and B. A. Wingate. *J. of Fluid Mechanics*, Vol. 757, 155–178 (2014).
- (29) *A rigorous bound on the vertical transport of heat in Rayleigh-Bénard convection at infinite Prandtl number with mixed thermal boundary conditions.* JPW and R. W. Wittenberg. *J. of Math. Phys.*, Vol. 55, 093104 (2014).
- (30) *Determining the effective resolution of advection schemes. Part II: Numerical Testing.* J. Kent, C. J. Jablonowski, JPW, and R. B. Rood. *J. Comp. Phys.*, Vol. 278, 497–508 (2014).
- (31) *Determining the effective resolution of advection schemes. Part I: Dispersion analysis.* J. Kent, JPW, C. J. Jablonowski, and R. B. Rood. *J. Comp. Phys.*, Vol. 278, 485–496 (2014).
- (32) *A bound on the vertical transport of heat in the ‘ultimate’ state of slippery convection at large Prandtl numbers.* X. Wang and JPW. *J. Fluid Mechanics*, Vol. 729, 103–122 (2013).
- (33) *Downscale Cascades in Tracer Transport Test Cases: An Intercomparison of the Dynamical Cores in the Community Atmosphere Model.* J. Kent, C. Jablonowski, JPW, and R. B. Rood. *Geo. Mod. Dev.*, Vol. 5, 1517–1530 (2012).
- (34) *Rigid bounds on heat transport by a fluid between slippery boundaries.* JPW and C. R. Doering, *J. Fluid Mechanics*, Vol. 707, 241–259 (2012).
- (35) *Assessing tracer transport algorithms and the impact of vertical resolution in a finite-volume dynamical core.* J. Kent, C. Jablonowski, JPW, and R. B. Rood. *Mon. Weath. Rev.*, Vol. 140, 1620–1638 (2012).
- (36) *The ultimate regime of two-dimensional Rayleigh-Bénard convection between free-slip fixed-temperature boundaries.* JPW and C. R. Doering, *Phys. Rev. Lett.*, Vol. 106, 244501, (2011).
- (37) *Internal heating driven convection at infinite Prandtl number.* JPW and C. R. Doering, *J. Math. Phys.*, Vol. 52, 093101 (2011).
- (38) *A stability analysis of divergence damping on a latitude-longitude grid.* JPW, C. Jablonowski, R. B. Rood, and P. H. Lauritzen, *Mon. Weath. Rev.*, Vol. 139, No. 9, 2976–2993 (2011).
- (39) *Asymptotic values, pre-poles, and periodic points.* JPW and L. Bakker, *Int. J. Bif. Chaos*, Vol. 20, No. 4 (2010).

Other Publications

Spontaneous formation of columnar vortices (November 2012) APS Gallery of Fluid Motion, Fluid Dynamics Video for APS DFD 2012 meeting (<http://arxiv.org/abs/1210.3620>).

Topics in geophysical fluid dynamics (Jan. 2012) Phd Dissertation, University of Michigan.

Topological bifurcations of Julia sets (April 2006) Brigham Young University Honors Thesis.

INVITED PRESENTATIONS (SINCE 2018)

- Bandung Institute of Technology (ITB) special guest lecture, Bandung, Indonesia (May 2025).
- Mathematics Department special seminar, National University of Singapore, Singapore (April 2025).
- BRIN (Indonesian National Research Agency) Lab-wide seminar, Bandung, Indonesia (April 2025).
- Uncertainty Quantification Lab-wide seminar, Sandia National Laboratory, virtual (February 2025).
- AGU Annual Meeting, Washington DC (December 2024).
- APS DFD, Salt Lake City, UT (November 2024).
- University of Indiana Statistics Colloquium, Bloomington, IN (September 2024).
- Brigham Young University Statistics Graduate Seminar, Provo, UT (September 2024).
- SIAM Annual Meeting, Spokane, WA (July 2024).
- AMS Southeast Sectional Meeting, Tallahassee, FL (March 2024).
- SIAM conference on Uncertainty Quantification, Trieste, Italy (February 2024).
- American Geophysical Union (AGU), San Francisco, CA (December 2023).
- Center for the Study of Complex Systems, University of Michigan (September 2023).
- International Congress of Industrial and Applied Mathematics (ICIAM), Tokyo, Japan (August 2023).
- Mathematics Department Colloquium, BYU-Hawaii (March 2023).
- Mathematics Department Colloquium, Utah State University (November 2022).
- Mathematics Department Colloquium, Southern Utah University (September 2022).
- Applied Math Geophysical/Astrophysical Fluids Seminar, University of Colorado Boulder (March 2022).
- AMS sectional meeting, Bloomington, Indiana (March 2022).
- SIAM conference on Analysis of PDEs (online only, March 2022).
- Theoretical Physics Seminar, Brigham Young University (March 2022).
- Geology Colloquium, Brigham Young University (October 2021).
- PDE seminar, University of Nebraska Department of Mathematics (March 2021).
- Mathematics Department Colloquium, Florida State University (February 2021).
- Joint Mathematics Meetings (online only, January 2021).
- SIAM Mathematics of Data Science Conference, held online (May 2020).
- Phase space analysis in complex systems - From quantum dynamics to turbulent flows: A Conference in Honor and in Memoriam of Bruno Eckhardt, Dresden, Germany (April 2020—Canceled due to COVID-19).
- SIAM Analysis of Partial Differential Equations, Palm Springs, CA, USA (December 2019).
- SIAM Application of Dynamical Systems Meeting, Snowbird, Utah, USA (May 2019).
- American Mathematical Society Sectional Meeting, Little Rock, Arkansas, USA (November 2018).

- American Mathematical Society Sectional Meeting, Ann Arbor, Michigan, USA (October 2018).
- Complex Systems Seminar, Universität Marburg, Marburg, Germany (May 2018).
- Applied and Computational Mathematics Seminar, Tulane University, New Orleans, LA, USA (March 2018).

POSTERS SINCE 2018

- *Bayesian methods for inferring magnitude and location of historical earthquakes*, AGU Annual Meeting, San Francisco, CA, USA (December 2023).
- *Applied data assimilation and parameter tracking*, AGU Annual Meeting, San Francisco, CA, USA (December 2023).
- *Data assimilation, parameter recovery, and equation discovery using a modified 'nudging' scheme*, AGU Annual Meeting, Chicago, IL, USA (December 2022).

CONFERENCE ORGANIZATION

- Institute for Foundations of Data Science workshop on Data Assimilation and Intersections with Machine Learning. Madison, WI, USA (August 2025).
- APS DFD local organizing committee (November 2024).
- Symposium commemorating the life and accomplishments of Charles R. Doering. Ann Arbor, MI, USA (May 2022).
- Rocky Mountain Partial Differential Equations: May 2017 in Provo, UT, USA.
- Challenges in Nonequilibrium Statistical Mechanics and Fluid Dynamics: A Conference in Honor of the 60th birthday of Charlie Doering. Co-PI, Provo, UT, USA (May 2016).

MINI-SYMPOSIA ORGANIZED

- *Data assimilation, system identification, and model discovery*, (20 speakers). 15th AIMS conference, Athens, Greece (July 2026).
- *UQ in data assimilation*, (4 speakers). SIAM Uncertainty Quantification, Minneapolis, MN, USA (March 2026).
- *Double Descent*, (8 speakers). SIAM Annual Meeting, Montreal Quebec, Canada (August 2025).
- *Mostly Bayesian approaches for Inverse Problems*, (12 speakers). SIAM Uncertainty Quantification, Trieste, Italy (February 2024).
- *Uncertainty Quantification in Modern Inverse Problems*, (8 speakers). SIAM Northern States Sectional Meeting, Logan, UT, USA (April 2023).
- *PDEs, Data, and Inverse Problems*, (20 speakers). AMS Western Sectional Meeting, Salt Lake City, UT, USA (October 2022).
- *PDEs, Data, and Inverse Problems*, (20 speakers). AMS Western Sectional Meeting (online only, October 2020).
- *Large-time dynamics of the Navier-Stokes equations and related PDEs*, (12 speakers, co-organized with A. Larios). SIAM Analysis of Partial Differential Equations, Scottsdale, AZ, USA (December 2015).
- *Wave-turbulence interactions in Geophysical and Astrophysical Fluid Dynamics*, (4 speakers, co-organized with I. Grooms). SIAM Application of Dynamical Systems, Snowbird, UT, USA (May 2015).

- *Regularity Problems for Nonlinear Partial Differential Equations Modeling Fluids and Complex Fluids*, (20 speakers, co-organized J. Bedroissian, T. K. Wong, and H. Jia). Joint Mathematics Meetings, Baltimore, MD, USA (January 2014).
- *The use of test cases in the development and improvement of dynamical cores*, (4 speakers, co-organized with J. Kent). SIAM Conference on Mathematical and Computational Issues in the Geosciences, Padua, Italy (June 2013).
- *Numerical investigations of convection at high Rayleigh numbers*, (4 speakers, co-organized with D. Goluskin). SIAM Conference on Mathematical and Computational Issues in the Geosciences, Padua, Italy (June 2013).
- *Separation of Scales in Geophysical Fluid Dynamics*, (4 speakers, co-organized with T. Haut). SIAM Conference on the Application of Dynamical Systems, Snowbird, UT, USA (May 2013).

STUDENTS SUPERVISED

Undergraduate:

Shane McQuarrie (2014–2016), Jessica Eggleston (Layton) (2014–2016), Austin Talbot (2014–2015), Tucker Albers (2014–2015), Marcus Horton (2014–2015), Ned Thomas (2014–2015), Allayna McFarlane (2015–2015, MS), Aaron Escamilla (2015–2016), David Reber (2015–2016), Webster Wong (2015–2018), Alexander White (2016–2017), Martha Morris (2017–2018), Spencer Giddens (2017–2018), Joshua Fullwood (2017–2019), Easton Dunn (2017–2017), Hunter Klein (Mechanical Engineering, 2017–2019), Bryce Berrett (Geology/Civil Engineering, 2017–2019), Cody Kessler (2018–2019), Adam Robertson (2018–2020), Garrett Carver (2018–present), McKay Harward (2018–2020), Josh Lapicola (2019–2021), Camille Carter (2019–2020), Ashley Avery (2019–present), Alice Oveson (2019–2019), Ryan Hilton (2019–2021), Mingyan Zhao (2019–2019), Kameron Lightheart (2020), Isaac Sorenson (2020–2022), Zac Yauney (2020–2021), Patrick Beal (2021), Dallin Stuart (2022–2024), Paul Smith (2022–present), Luke Green (2023), Ilha Hwang (2023–present), Ashley Spencer (2023–present), Nephi Suryama (2024–present), Cameron Chamberlain (2024), Wyatt Wimmer (2023–present), Ashtyn Whipple (2024–present), Addison Dhulst (2024–2025), Benjamin Harris (2024), Brie Barzee (2024), Emily Martin (2024), Stella Martin (2024), Hannah Bloomfield (2024–2025), Thea Spigarelli (2024–present), Norman Rasmussen (2024–present), Rowan Williams (2025–present), Emmaline McKinnon (2025–present), Jacob Schonlau (2026–present), Quinn Oveson (2026–present), Spencer Bringham (2026–present)

Masters (chair only):

Allayna McFarlane (2015–2017), Shane McQuarrie (2016–2018), Yajing Zhao (2017–2020), Spencer Giddens (2018–2020), Hayden Ringer (2018–2020), Benjamin Pachev (2019–2020), McKenna Pitts (2020–2021), Marja Crawford (2019–2022), Kameron Lightheart (2020–2021), Taylor Paskett (2020–2022), James Griffin (2020–2022), Carol Herrera Glenn (2021–2022), Jake Callahan (2021–2023), Raelynn Wonnacott (2021–2023), Chelsey Noorda (2021–2023), Kelly Chang (2022–2024), Joshua Newey (2022–2024), Collin Free (2022–2023), Paul Smith (2023–2025), Nathan Schill (2024–present), Melanie Neller (2024–present), Lydia Tolman (2025–present), Tiara Eddington (2025–present), Eli Sampson (2025–present), Wyatt Wimmer (2025–present).

PhD:

Zhao Pan (Mechanical Engineering, 2015–2016), Ryan Wood (Chemical Engineering 2017–2019)

Served as a committee member for over 50 other graduate students (not including those listed above).

ACADEMIC HONORS, GRANTS, ETC.

- *Collaborative Research: Augmenting Continuous Data Assimilation to Perform Equation Discovery with Applications in Geophysics*, \$132K: National Science Foundation.
- *Inclusion of Shaking into Bayesian inversion of Paleo-seismic events in Indonesia*, \$25K: College High Impact Research Proposal (CHIRP, internal BYU funds). 2024-2027.
- *XTRIPODS: Advances and novel applications of closed-loop data assimilation, and educational improvements to data science courses*, National Science Foundation, \$199,878, 2024-2026.
- *Fulbright Scholar Award: Bandung, Indonesia*, Fulbright Foundation, State Department. Approximately \$30K, 2025.
- *Distinguished Mentoring Award*, BYU Mathematics Department. 2023.
- *Mentoring graduate students in the estimation and uncertainty quantification of archaeological seismic problems*, \$20K: College High Impact Research Proposal (CHIRP, internal BYU funds). 2023-2025.
- *Data Assimilation for Turbulent Flows: Dynamic Model Learning and Solution Capturing*, National Science Foundation, \$174,703. 2022-2025.
- *Distinguished Teaching Award*, BYU Mathematics Department. 2020.
- *Using shaking intensity observations to improve estimation of historical seismic events*, \$15K: Graduate student mentoring grant (internal BYU funds). 2019–2020.
- *Distinguished Research Award*, BYU Mathematics Department. 2018.
- *Analysis and Computation of Geophysical Fluid Dynamics*, \$42K: Simons Foundation Collaboration Grant. 2018-2023.
- *Quantifying Historical Indonesian Seismicity via Bayesian Statistical Inversion*, \$20K: Mentored Environment Grant (internal BYU funds). 2018–2019.
- *Rocky Mountain PDEs*, \$17,875, co-PI with M. Allen and B. Barker (BYU), and G. Richards (Utah State): NSF Conference Grant. 2017.
- *Interdisciplinary Mentoring, Integrating the Study of Network Dynamics, Control Theory, and Robotics*, \$20K, co-PI with B. Webb and M. Killpack (BYU): Mentored Environment Grant (internal BYU funds). 2016–2018.
- *Bridge over troubled waters*, \$100K, co-PI with R. Harris and C. Emmitt (BYU): Geoscientists without Borders. 2016–2019.
- *National Science Foundation Postdoctoral Research Fellowship*, Florida State University (X. Wang and M. Gunzburger) –declined for postdoc position at LANL.
- *Asymptotic scaling of heat transport in infinite Prandtl fluids at very high resolution*, co-PI with H. Johnston and B. K. Muite, National Science Foundation Teragrid (XSEDE) award TG-CTS 110010.
- 2nd place Michigan Geophysical Union Poster competition, March 2010.
- *Orson Pratt Award*, April 2005 (Outstanding Senior in Mathematics Department, BYU).

JOURNALS REFEREED FOR:

Physics Letters A, Proceedings of the Royal Society A, Journal of Mathematical Physics,

Journal of Mathematical Analysis and Applications, Communications in Mathematical Physics, Physical Review E, Physical Review Letters, Physical Review Fluids, Physics of Fluids, Nonlinearity, Journal of Fluid Mechanics, Quarterly Journal of the Royal Meteorological Society, Journal of Climate, ASME Journal of Fluids Engineering, Journal of Fluid Mechanics RAPIDS, Physica D., Geophys. J. Int., Zeitschrift für Angewandte Mathematik und Mechanik, New Zealand Journal of Geology and Geophysics

MEMBERSHIP IN PROFESSIONAL ORGANIZATIONS:

Society for Industrial and Applied Mathematics (SIAM), American Physical Society (APS), American Mathematical Society (AMS), American Geophysical Union (AGU).

EDITORIAL BOARDS:

Chief Guest Editor for a special issue in Physica D in memoriam of Charles R. Doering.